VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

Application ID: [Enter ID]

VergeLink — camera-free eye + jaw-clench control of fan and light for people with quadriplegia

Team ID: [Team ID] | Track: [Track] | Dept. of Biomedical Engineering, VCET Erode (Batch 23BM7A)

Faculty guide: Mrs. C. Radhika · Industry mentor: K. P. Amalnath, Mediowel Healthineers

Problem Context & User Journey: Before State

~20,000 / yr
NEW SPINAL CORD INJURIES, INDIA

1.5 million
INDIANS LIVING WITH SCI

54.4 lakh
MOVEMENT DISABILITY (CENSUS 2011)

~12.9 h / day
UNPAID CARE, SCI USERS (US STUDY)



Figure: Before-state user flow for a single fan or light change.

Problem Statement

People with quadriplegia who can still move their eyes and jaw cannot switch their own fan or light. Every change waits on a caregiver, and the hands-free options on the market depend on a camera, on clear speech, or cost well beyond most Indian households.

Target Beneficiary & Before-State User Flow

- **Project overview:** VergeLink is a camera-free wearable that lets a person switch their own fan and light by looking at a target and clenching the jaw once.
- **Target user:** Adults with high cervical spinal cord injury, ALS/MND or similar conditions who keep eye movement and jaw control, and the family caregivers who look after them at home or on rehab wards.
- **Operational bottleneck:** Every environmental change needs another person. The user calls out (if voice allows), the caregiver stops their own work, walks over and toggles the switch, then repeats this day and night.
- **Public search disclosure:** Camera eye-trackers need steady lighting and a screen-facing camera. The Indian EOG communicator Netravaad is priced at ₹1.65 lakh. VergeIO (2025) showed bilateral EOG can tell 4 depth gestures apart, but as a glasses input study, not home-appliance control for paralysed users.

Core innovation: Bilateral EOG vergence + ToF depth evidence + a deliberate jaw-clench (EMG) confirmation gives two-factor, camera-free appliance control built from ₹13,093 of ordered parts. Team ID: [Team ID] | Track: [Track]

User Journey: After State & Quantified Efficiency

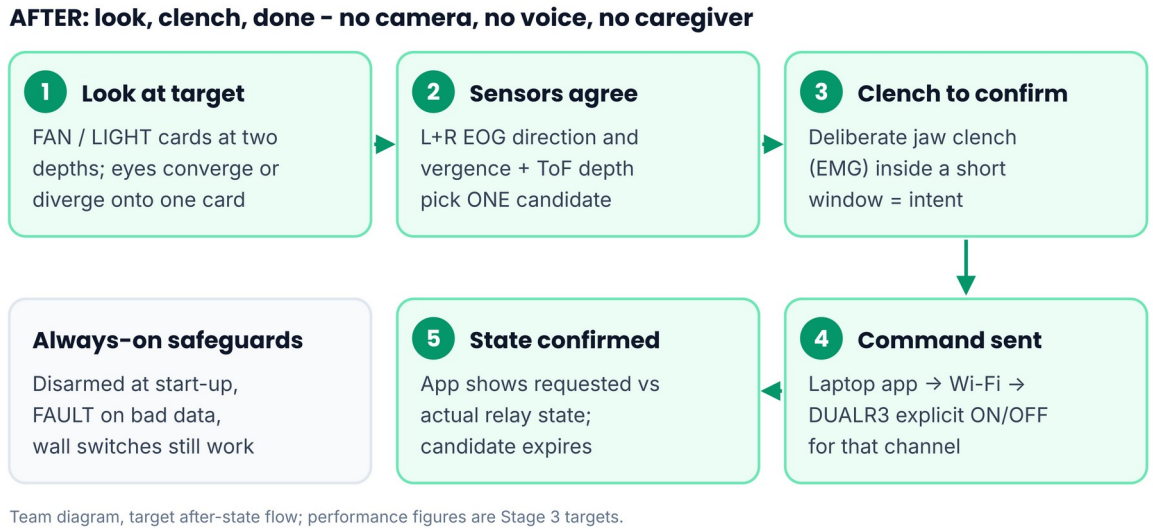


Figure: Target after-state flow — look, clench, relay switches.

How these targets will be measured (Stage 3)

- **Timed trials:** app logs time from target shown to relay state confirmed, per attempt.
- **False activations:** 30-minute sessions of rest, talking, chewing and blinking with the system armed.
- **Accuracy:** 2 targets × 2 depths, scripted order; first with team volunteers under institutional approval.

Before vs VergeLink (targets)

Metric	Before	VergeLink target
Who switches	Caregiver	User alone
Time per change	Caregiver response time	≤ 3 s look → clench → relay
Hands-free cost	₹1.65 L (Netravaad)	₹13,093 ordered BOM (−92%)
Camera / mic	Camera or voice needed	Neither
False activations	n/a	≤ 1 per 30 min armed
Selection accuracy	n/a	≥ 90% (2 targets × 2 depths)
Caregiver trips	Every fan/light change	Zero for these two loads

Efficiency claim scope: only the cost row is a fact today (Netravaad price vs our recorded orders). Every other VergeLink figure is a design target; parts were ordered on 6 Oct 2026 and bench measurement starts after delivery checks.

Beneficiary impact: routine fan and light changes become self-directed, so the caregiver's time goes to care that genuinely needs a person — and the user is not left waiting in the heat or dark.

System Architecture & Specs

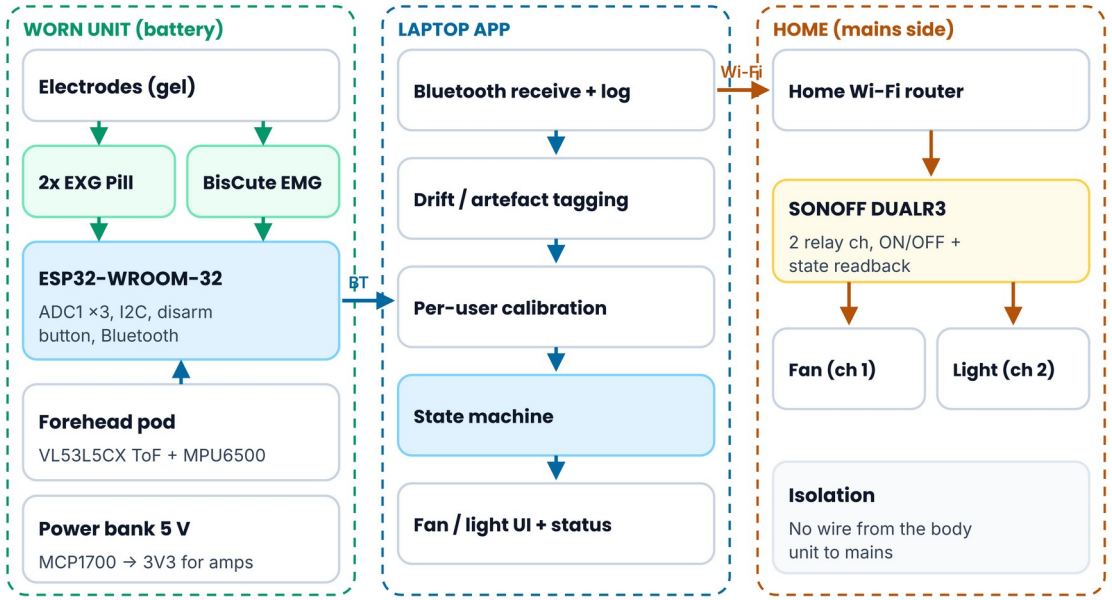


Figure: System block diagram — worn sensing unit, laptop application, and mains-side relay.

Firmware / logic flow

- ESP32 samples 3 analog channels + I2C sensors, timestamps every frame, flags clipping / stale / missing data.
- Laptop state machine: DISARMED → ARMED → CANDIDATE (gaze + depth agree) → CONFIRMED (clench in window) → ON/OFF.

Engineering Target Specifications & Subsystems

Parameter	Target spec	Selected hardware / protocol
Eye signals	2 ch bilateral horizontal EOG	2× Upside Down Labs BioAmp EXG Pill
Intent confirm	1 ch jaw EMG (72–720 Hz band)	Muscle BioAmp BisCute
Depth evidence	8×8 zones, up to 4 m	ST VL53L5CX ToF, I2C
Head motion	6-axis, motion rejection	MPU6500 (GY-6500), I2C
Controller	3× ADC1, ≥ 500 Hz/ch	ESP32-WROOM-32 DevKit
Laptop link	< 50 ms frame latency	Bluetooth Classic serial
Appliances	2 ch, explicit ON / OFF	SONOFF DUALR3 via home Wi-Fi
Power	≥ 8 h session	10,000 mAh bank; MCP1700 3V3_AFE

All rate, latency and runtime values are targets to be verified at bring-up.

Architecture rationale: analog inputs use ADC1 only because ESP32 ADC2 cannot be used while the radio runs; the bio-amps get their own MCP1700 rail; the body-worn unit has no wired path to mains — commands reach the relay only over Wi-Fi.

CAD Design & Mechanical Specs

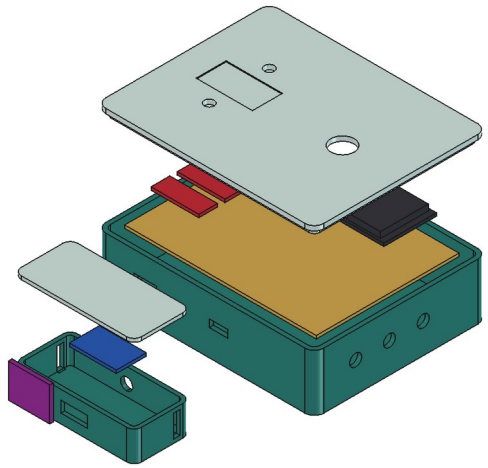
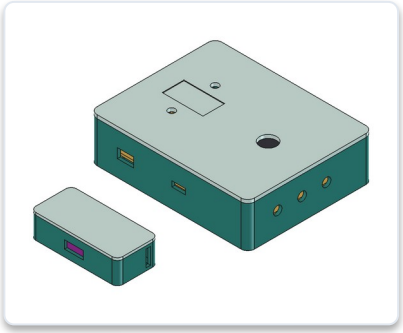
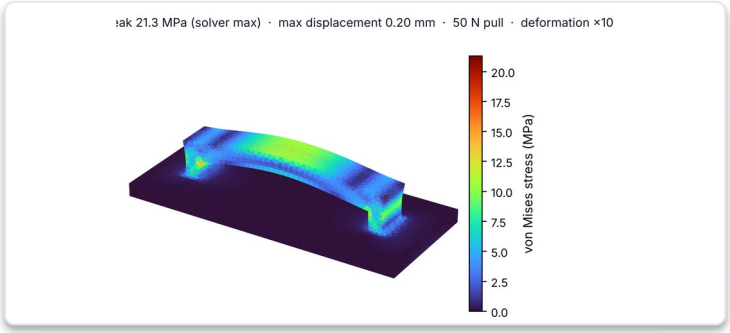


Figure: FreeCAD 1.1 exploded assembly — enclosure 116×92×30 mm, pod 64×30×16 mm.



Assembled view.



CalculiX FEA of the strap loop (von Mises).

Mechanical Validation & FEA Stress Report

- **Enclosure:** 2.2 mm PETG walls, 5 mm corner radii, four M2.5 standoffs on the 3×4 in perfboard pattern; ports for 3 electrode leads, pod cable, USB power and ESP32 programming; lid holes for the disarm button and 2 status LEDs.
- **Forehead pod:** 14×10 mm ToF window, IMU on an internal shelf, 26 mm slots for an elastic headband, rear cable exit.
- **Material selection:** PETG over PLA (PLA softens at modest temperatures near the body and in a warm room) and over ABS (warps on an open-frame printer); printed on the college FDM printer.
- **FEA load case:** strap loop (30×6×6 mm, 2 mm posts, 3 mm bridge) on a 2.2 mm wall patch; wall face fixed; 50 N pull on the bridge away from the case (strap snag). 44,420-node 2nd-order tet mesh, E = 2.1 GPa, v = 0.38.
- **Result:** peak von Mises 21.3 MPa at the sharp inner corner where bridge meets post; max displacement 0.20 mm. Against ~45–50 MPa typical PETG datasheet yield, FoS ≈ 2.1–2.3 (bulk).
- **Next action:** add a 1 mm fillet at that corner and print with layers along the load path, since FDM inter-layer strength is below bulk.

Design iteration: CAD review caught the strap-loop gap modelled on the wrong side (loop was solid against the case) — corrected before FEA. Native files: VergeLink_Enclosure.FCStd / .step, VergeLink_StrapLoop_FEA.FCStd.

Electronics Schematics & BOM

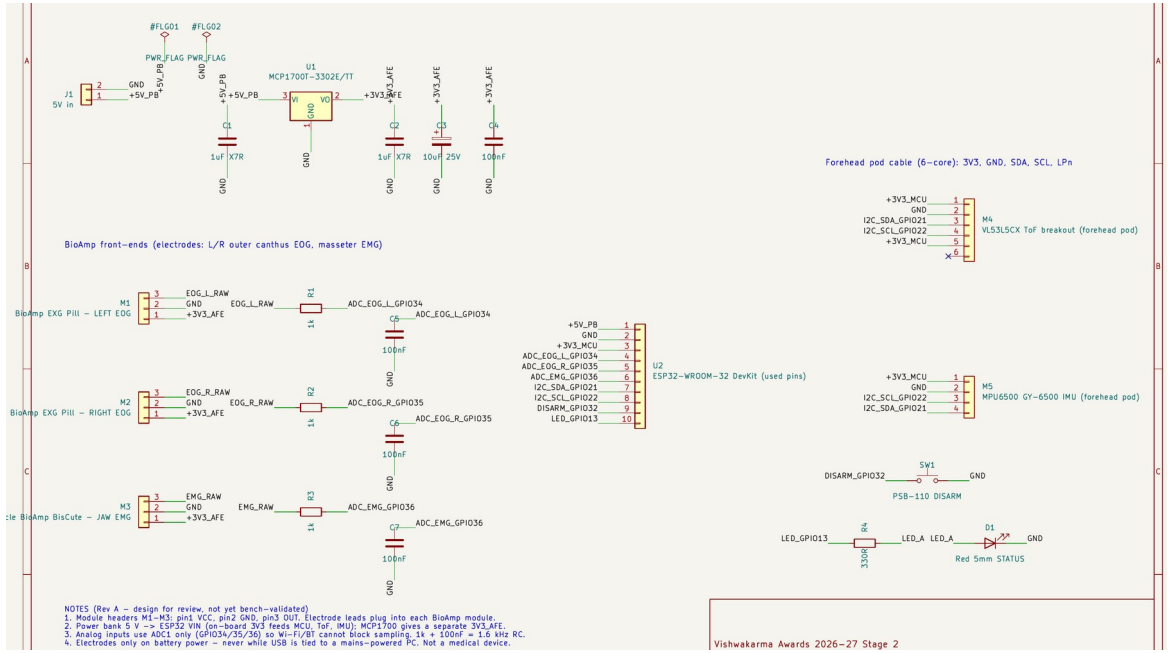


Figure: KiCad 10 schematic, Rev A — ERC 0 errors / 0 warnings (file VergeLink_Mainboard.kicad_sch).

Power distribution & regulation

- Power bank 5 V → ESP32 VIN (its on-board 3V3 feeds MCU, ToF, IMU). MCP1700 (250 mA LDO, 1 µF in/out) feeds only the three bio-amps.
- 1 kΩ + 100 nF RC (1.6 kHz) on each ADC input; rail currents measured at bring-up.

Itemized Prototype Bill of Materials (as ordered)

Component	Part / specification	Qty	Vendor (India)	Cost ₹
EOG front-end	BioAmp EXG Pill kit	2	Upside Down Labs	5,198.00
EMG front-end	BisCute kit + cable + electrodes	1	Upside Down Labs	1,346.00
Wi-Fi relay	SONOFF DUALR3 (2 ch)	1	SONOFF India	1,665.06
Power bank	Portronics Aero 10, 10,000 mAh	1	Portronics	1,303.00
Depth sensor	SmartElex VL53L5CX breakout	1	RoboTechMinds	1,096.80
MCU	ESP32-WROOM-32 38-pin DevKit	1	Robu	421.00
IMU	GY-6500 MPU6500	1	FlyRobo	345.00
LDO	MCP1700T-3302E/TT	2	Robu	145.00
Passives & build	Resistor kit, caps, LEDs, perfboard, wire, button, OLED, cables	lot	Robu / IHC / ElectronComponents	1,573.41
Total (incl. GST & shipping)				**13,093.27**

Approved cap ₹16,000 → headroom ₹2,906.73 (printing, targets and electrician allowances come from this).

Procurement status: every part ordered from Indian sellers by 6 Oct 2026 — no imports. Delivery inventory and module-revision checks are in progress, so no lead time is claimed as met.

Your Maker Journey

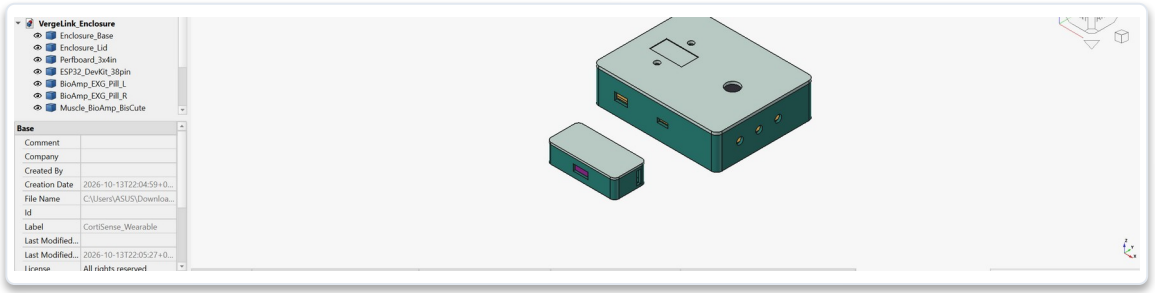


Figure: FreeCAD 1.1 session — VergeLink_Enclosure feature tree and live model (14 Oct 2026).

[Add photo] Team members soldering the perfboard / assembling the enclosure, wearing institutional ID badges, plus a joint review photo with the faculty guide.

Stage 1 to Stage 2 Engineering Iterations

- **Front-end pivot:** Stage 1 planned ADS1293 + AD8232 + nRF52840. The ₹6,995 ADS1293 order was cancelled; two BioAmp EXG Pills + BisCute + ESP32 keep the build inside ₹16,000 and match the bilateral EOG set-up used in VergeIO (2025).
- **Scope made honest:** third (vertical) EOG channel, magnetometer heading, wheelchair DAC and dry electrodes are out of this build; Stage 1 simulation figures are not reported as results.
- **Actuation:** two separate MINIR4 relays replaced by one SONOFF DUALR3 — two channels in one retrofit box.
- **CAD milestone:** 12-part FreeCAD assembly, STEP export and a CalculiX FEA of the strap loop completed on 14 Oct 2026.
- **EDA milestone:** KiCad 10 schematic with 21 components and 15 named nets; ERC clean (0 errors, 0 warnings) on 14 Oct 2026; netlist and BOM exported.
- **Raw repository:** github.com/amalnathkp13-boop/vergelink — native .FCStd, .step, .kicad_sch, netlist, ERC report, BOM, figures and this deck.

Institutional sign-off: Faculty guide Mrs. C. Radhika, Dept. of Biomedical Engineering, VCET Erode (Batch 23BM7A) — [add review date / signature].

Safety & Hazard Mitigation

FMEA Matrix — failure modes and embedded protections

Failure mode	Potential hazard	Embedded fail-safe / mitigation response
Electrodes worn while USB goes to a mains PC	Leakage current through skin electrodes	Battery-only operation whenever electrodes are attached; program over USB only with electrodes removed; no wired path to the relay.
Chewing / talking produces jaw EMG	Fan or light switches unintentionally	EMG accepted only inside a candidate window after gaze + depth agree; refractory period; physical disarm button (GPIO32).
EOG drift, electrode peel, sensor dropout	Wrong target selected	Clipping / stale / missing-sensor flags force FAULT; per-session calibration; candidates expire automatically.
Bluetooth or Wi-Fi link drops	Command lost or repeated	Restart in DISARMED; explicit ON / OFF (never toggle); app shows requested vs actual relay state.
Supply sag / LDO overload	Noisy or clipped bio-signals	Bio-amps on a separate MCP1700 rail; ESP32 and radio on the DevKit regulator; rail currents measured at bring-up.
Mains wiring of the DUALR3	Shock or fire	Licensed electrician installs in existing switchboard; wall switch and fan regulator stay in service.

Physical fail-safes

- **Disarm button:** momentary PSB-110 on GPIO32 drops the system to DISARMED at once.
- **Enclosures:** PETG case and pod cover all boards; no exposed conductors near the skin.
- **Isolation:** body-worn unit and mains relay share no wire — only Wi-Fi.

Standards Compliance (considered, not certified)

- **ISO 14971:** risk-management file started from this FMEA.
- **IEC 60601-1:** leakage-current principles for body-worn electrodes.
- **IEC 60825-1:** VL53L5CX emitter is Class 1 (per ST).

Human testing: electrode trials only on team members first, and volunteer sessions only after institutional ethics approval. No conformity or medical-device claim is made.

Roadmap: 4-Month Fabrication Plan

M1

Oct 2026 · Fabrication

- Delivery inventory and module revision checks
- Perfboard soldering per Rev A schematic
- ADC bring-up with a signal generator — no body contact yet

M2

Nov 2026 · Fabrication

- PETG print of enclosure + pod on the college printer
- Firmware: timed acquisition + health flags
- Laptop app v0; electrode trials on team members

M3

Dec 2026 · Integration

- Calibration routine and two-factor state machine
- ToF depth gating, IMU motion rejection
- DUALR3 fitted by an electrician and integrated

M4

Jan 2027 · Field demo

- Timed trials: accuracy, false triggers, latency
- Stage 2 → 3 report with measured data
- Demo rehearsal with the faculty guide

Stage 3 Target

- **Commitment:** a working prototype that switches a real fan and light from look + clench, ready for functional testing in Stage 3.
- **Evidence we will bring:** logged accuracy, false-activation and latency numbers from scripted sessions, plus the commits and photos that show each milestone.

Dependencies: parts delivery (ordered by 6 Oct 2026), college 3D-printer slot, electrician visit for the DUALR3, and ethics approval before any volunteer testing.

Technical Challenges & Risks

Bottleneck: EOG drift vs vergence

- **Constraint:** The near/far vergence change is small next to slow electrode drift. A strong high-pass filter would remove the drift — and the vergence signal with it.
- **If unmanaged:** Wrong depth chosen; selections that wander over a session.

Bottleneck: ESP32 ADC quality

- **Constraint:** The ESP32 ADC is non-linear near its range limits and noisy, and ADC2 is unavailable while Wi-Fi or Bluetooth runs.
- **If unmanaged:** Distorted EOG / EMG traces; poor repeatability between units.

Bottleneck: false jaw-clench triggers

- **Constraint:** Chewing, talking and swallowing all produce masseter EMG that can look like a deliberate clench.
- **If unmanaged:** Unwanted fan or light switching during meals or conversation.

Other risks tracked

- **ToF on dark / glossy targets:** measured surface distance may not match where the eyes converge.
- **No magnetometer:** MPU6500 yaw drifts, so it cannot give a stable room heading.
- **Relay control path:** the DUALR3's local (LAN) vs cloud control options must be confirmed before the app is written.

Why this matters: each of these could make the demo look like it works on the bench but fail with a real user — so each one has a planned test on the next slide.

Potential Solutions & Risk Mitigation

Risk	Primary mitigation	Backup / fallback	How we will verify
EOG drift vs vergence	Keep low-frequency path; per-trial baseline removal; discrete near/far targets (as in VergeIO)	Require ToF depth agreement before any candidate	10-min fixed-gaze recordings at both depths
ESP32 ADC quality	ADC1 only, linear input band, 1.6 kHz RC, oversampling, Espressif calibration API	External I2C ADC from the ₹2.9k headroom	Signal-generator sine sweep before body contact
False clench triggers	Clench must be held inside a candidate window; per-user threshold; refractory period	Raise threshold / add dwell; physical disarm	Scripted chew / talk / rest sessions, false triggers counted
ToF on dark / glossy targets	Matte, high-contrast target cards	Use ToF only as a gate, not as gaze depth	Distance tests on matte, dark and glossy cards
IMU yaw drift	Use IMU only to reject head motion	Fixed-head demo posture	Logged yaw during 10-min sessions
Relay control path	Confirm DUALR3 local-LAN control first	Vendor app / Home Assistant bridge	Explicit ON/OFF + state readback test

Verification plan: every risk above has a bench test; results — including failures — will be reported with the Stage 3 submission.

References & Sources

[1] Zhang X. et al., “VergeIO: Depth-Aware Eye Interaction on Glasses,” arXiv:2507.02187 (2025) — arxiv.org/abs/2507.02187

[2] Upside Down Labs, BioAmp EXG Pill documentation — docs.upsidedownlabs.tech/hardware/bioamp/bioamp-exg-pill

[3] Upside Down Labs, Muscle BioAmp BisCute documentation — docs.upsidedownlabs.tech/hardware/bioamp/muscle-bioamp-biscute

[4] STMicroelectronics, VL53L5CX Time-of-Flight 8×8 multizone sensor — st.com/en/imaging-and-photonics-solutions/vl53l5cx.html

[5] Espressif, ESP-IDF ADC (oneshot) driver and calibration documentation, ESP32 — docs.espressif.com

[6] Microchip, MCP1700 low-quiescent-current LDO datasheet (DS20001826)

[7] SONOFF DUALR3 two-gang smart switch — sonoff.in

[8] International Health Policies, “Spinal cord injuries in India: towards a comprehensive preventive and rehabilitative response”

[9] Office of the Registrar General & Census Commissioner, India — Census 2011 disability data (via punarbhava.in)

[10] SCIRE Project, Non-intervention attendant care studies (Robinson-Whelan & Rintala 2003) — scireproject.com

[11] Amrita Vishwa Vidyapeetham, news release on Netravaad eye-gesture communication device — amrita.edu

Data sources: statistics on slide 2 and the Netravaad price on slides 2–3 come from references [8]–[11]; component specifications from [2]–[7].

Image Sources

Slide	Figure	Source / tool
2	Before-state user flow	Project diagram (drawn for this submission)
3	After-state user flow	Project diagram (drawn for this submission)
4	System block diagram	Project diagram (drawn for this submission)
5	Exploded and assembled enclosure views	FreeCAD 1.1 — cad/VergeLink_Enclosure.FCStd
5	Strap-loop von Mises stress plot	FreeCAD FEM + CalculiX — cad/VergeLink_StrapLoop_FEA.FCStd
6	Mainboard schematic	KiCad 10 — kicad/VergeLink_Mainboard.kicad_sch
7	FreeCAD session screenshot	Screen capture of the project model

Third-party images: none. All files are in the project repository: github.com/amalnathkp13-boop/vergelink